

User Manual

Final Project: Simon Say What

Multi-Player Memory Game on FPGA

ECE 6370 Advanced Digital Design

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1. System Description

This system is a single-player hex-sequence memory game (Simon-Says variant) running on the Terasic DE0-CV FPGA board. The objective is to memorize and reproduce a sequence of randomly generated hexadecimal digits within a 99-second round. Score accumulates with each correct entry, weighted by the difficulty level chosen at the start of the round.

Before the game can be played, the system must authenticate the player using a 4-digit ID and a 4-digit password. Both are stored on-chip in dedicated ROMs. Until the correct ID and password are entered, the toggle switches and player keys have no effect on gameplay. Once authenticated, the player selects a difficulty level (easy, medium, or hard), presses the Game Start button, and a 99-second timer begins counting down on the leftmost two 7-segment displays.

During gameplay, the system displays a 4-bit random hex digit on the rightmost 7-segment display (HEX0) for a difficulty-paced interval. The player then re-enters that digit on the right four switches (SW[3:0]) and confirms with the player load button (KEY[1]). Each correct entry increments the score by the difficulty multiplier (1x, 2x, or 4x). The score is shown live during the round.

When the timer reaches zero, the round ends and the round score is committed to BRAM. The system enters the leaderboard view, where the player can toggle between their personal best and the all-time global best, restart the game with the same difficulty, or log out and return to the authentication screen.

2. Board Layout

The diagram below shows the DE0-CV board layout. The labeled controls and displays in this manual are referenced against this figure.

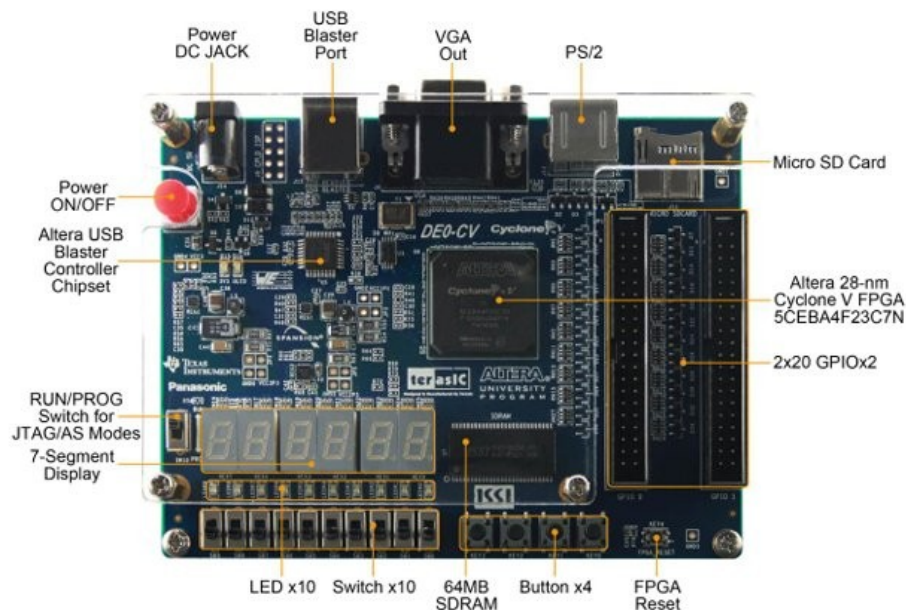


Figure 1 — DE0-CV board with controls and displays used by the Hex Sequence game.

Table 1: Board Controls Reference

Control	Function
SW[9:6] — 4 left switches	ID / password digit entry (during authentication)
SW[5:4] — difficulty switches	Difficulty select: 00 = none, 01 = easy, 10 = medium, 11 = hard
SW[3:0] — 4 right switches	Player digit input during gameplay
KEY[0]	RESET — clears the system and locks it back to UNAUTHENTICATED
KEY[1]	Player Load / Leaderboard Scroll
KEY[2]	ID/PW digit Enter / Game Start / Round Restart
KEY[3]	Logout button (active only in WAIT_FOR_START and LBOARD)
HEX0	Random hex digit during gameplay; least-significant score nibble during leaderboard
HEX1	Score and leaderboard digit
HEX2 – HEX3	Leaderboard NumberID digits
HEX4 – HEX5	99-second countdown timer (tens and ones)
LEDR[0]	UNAUTHENTICATED — entering ID
LEDR[1]	ID_IN — ID accepted, entering password
LEDR[2]	AUTHENTICATED — single-cycle flash (typically not visible)
LEDR[3]	WAIT_FOR_START — select difficulty and press start
LEDR[4]	GAME_RUNNING — round active
LEDR[5]	END_GAME — 1-second hold while score is written to BRAM
LEDR[6]	LBOARD — leaderboard view
LEDR[9:7]	Unused (tied LOW in production)

	Button input			
Meeting discussed	Key 1	Key 2	Key 3	Key 4
DE0-CV	Key 3	Key 2	Key 1	Key 0
Functionality				
User Authntication	ID/Password Entry			Rest
Logged in / Difficutly Selection	Log Out		Set Difficulty	Rest
Game Start	Log Out	Game Start		Rest
Game Running			Entry Number	Rest
Round Finished / waiting / Difficulty state	Log Out		Toggle Leaderboard	Rest

Key layout.

3. How to Authenticate the System

The system must be authenticated before the player can start a round. Authentication is a two-stage process: first the player enters a 4-digit ID, then a 4-digit password. Both are submitted one digit at a time on the left four switches using KEY[2] to confirm each digit. Each successful entry advances the system to the next state, lighting the corresponding phase LED on LEDR[1:0].

Valid ID / Password Pairs

The system stores four valid player accounts. Use any of the following ID / password pairs to log in:

Player Index	ID (hex)	Password (hex)
0	1234	5810
1	5678	4525
2	ABCD	3998
3	CDEF	1032

Step-by-Step Login

1. Power on the board or press KEY[0] (RESET). LEDR[0] (UNAUTHENTICATED) should be ON; all other phase LEDs OFF.
2. Using the left four switches (SW[9:6]), set the first ID digit in hex. For ID 1234, set SW[9:6] = 0001.
3. Press KEY[2] once to submit the digit.
4. Set the second ID digit (0010 for '2'), press KEY[2]. Repeat for the third (0011) and fourth (0100) digits.
5. If all four digits match a stored ID, LEDR[0] turns OFF and LEDR[1] (ID_IN) turns ON. The system is now in password-entry mode.
6. Repeat the four-digit entry process for the password. For ID 1234 the password is 5810, so enter SW[9:6] = 0101 → KEY[2], 1000 → KEY[2], 0001 → KEY[2], 0000 → KEY[2].
7. If the password matches, LEDR[1] turns OFF, LEDR[2] (AUTHENTICATED) flashes for one clock cycle, and LEDR[3] (WAIT_FOR_START) turns ON. The system is now authenticated and ready to start a round.

If you enter an incorrect digit, the authentication FSM stays on the current digit position and lets you retry. After four failed password attempts, the system resets back to ID entry (LEDR[0] ON).

4. How to Select Difficulty

With LEDR[3] (WAIT_FOR_START) ON, the player must select a difficulty level before the round can begin. Difficulty is set on SW[5:4]:

SW[5:4]	Difficulty	Per-digit display window	Score multiplier
00	None — round cannot start	—	—
01	Easy	20 s	× 1
10	Medium	15 s	× 2
11	Hard	10 s	× 4

The chosen difficulty is latched the moment KEY[2] is pressed to start the game. Once latched, changing SW[5:4] mid-round has no effect.

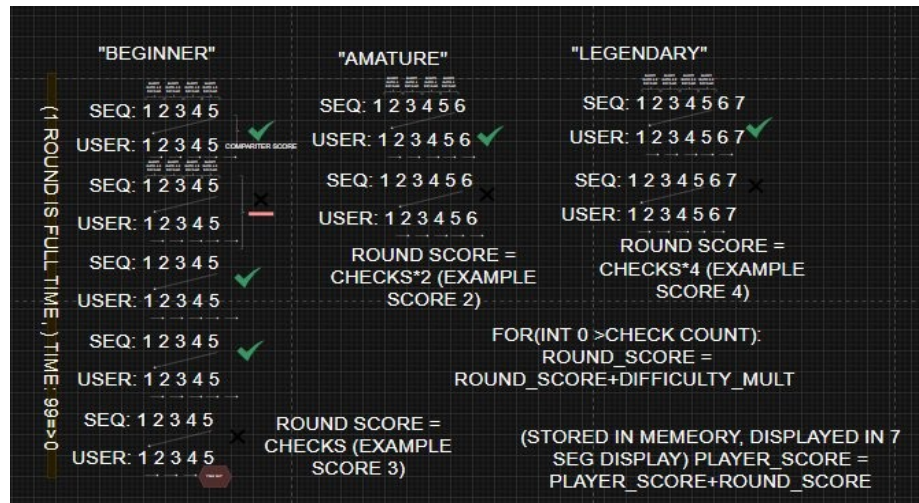


Figure 2 — Difficulty progression. Each tier increases the sequence length and the score multiplier per correct entry.

5. How to Play the Game

Starting the Round

1. With LEDR[3] ON and SW[5:4] set to a non-zero value, press KEY[2] once to start the 99-second round.
2. LEDR[3] turns OFF, LEDR[4] (GAME_RUNNING) turns ON. The countdown timer on HEX5–HEX4 begins decrementing from 99 toward 00 at one second per tick.

Memorize the Sequence

3. A random hex digit appears on HEX0 (the rightmost 7-segment display) for the duration of the difficulty window (20 s / 15 s / 10 s).
4. Memorize the displayed digit. The system will display a new random digit each window until the round ends.

Enter Your Match

5. Within the display window, set SW[3:0] to the 4-bit value matching the displayed digit.
6. Press KEY[1] (Player Load) to lock in your entry. If your entry matches the displayed digit, the score increments by the difficulty multiplier. If it does not match, no points are awarded for that window.

End of Round

7. When the timer reaches 00, LEDR[4] turns OFF and LEDR[5] (END_GAME) turns ON for one second.
8. During this hold the round score is written to BRAM. If your score is higher than your previous personal best for this account, the per-player record is updated. If it is higher than the global best across all accounts, the all-time leaderboard is updated.
9. LEDR[5] then turns OFF and LEDR[6] (LBOARD) turns ON. The leaderboard view opens with your personal best displayed.

6. Leaderboard, Restart, and Logout

LEDR[6] indicates that the system is in leaderboard view. Three actions are available from this state:

Button	Action
KEY[1]	Scroll — toggles between your personal best and the global best. The displayed entry shows '<NumberID> <score>' packed across HEX5–HEX0.
KEY[2]	Restart Round — reloads the timer to 99 and returns to WAIT_FOR_START (LEDR[3] ON) using the previously selected difficulty.
KEY[3]	Logout — returns the system to UNAUTHENTICATED (LEDR[0] ON). Any further play requires a fresh ID and password entry.

Note: KEY[3] is only sampled by the game controller while LEDR[3] (WAIT_FOR_START) or LEDR[6] (LBOARD) is ON. Pressing KEY[3] during gameplay or authentication has no effect, so accidental presses during a round will not log the player out.

Full Reset

At any time, pressing KEY[0] (RESET) clears the entire system, including the score registers and the in-progress authentication state, and returns the board to UNAUTHENTICATED with LEDR[0] ON. RESET does not erase the BRAM-stored leaderboard entries written during prior rounds in the same power session but does clear them on power cycle.